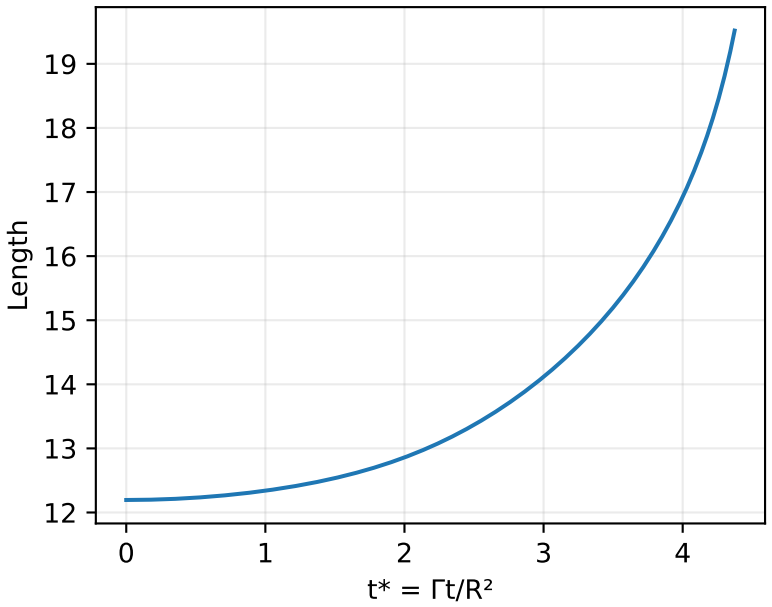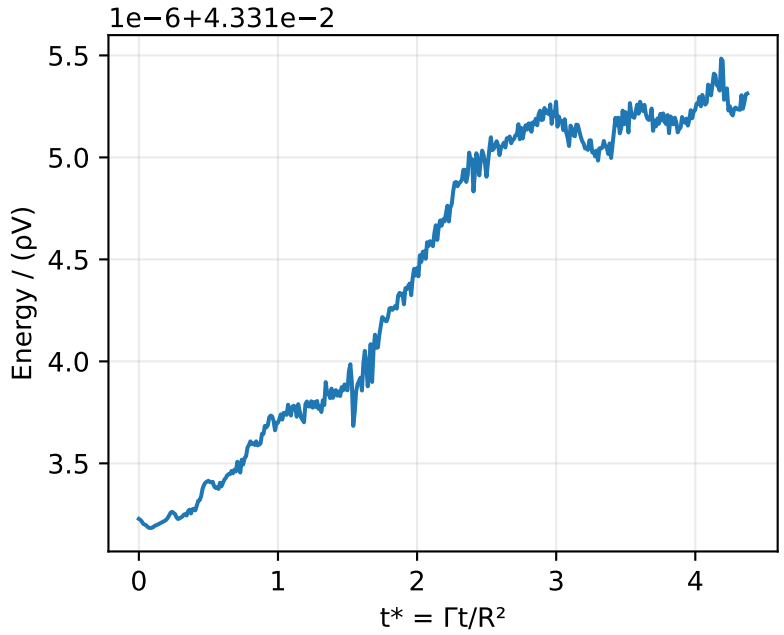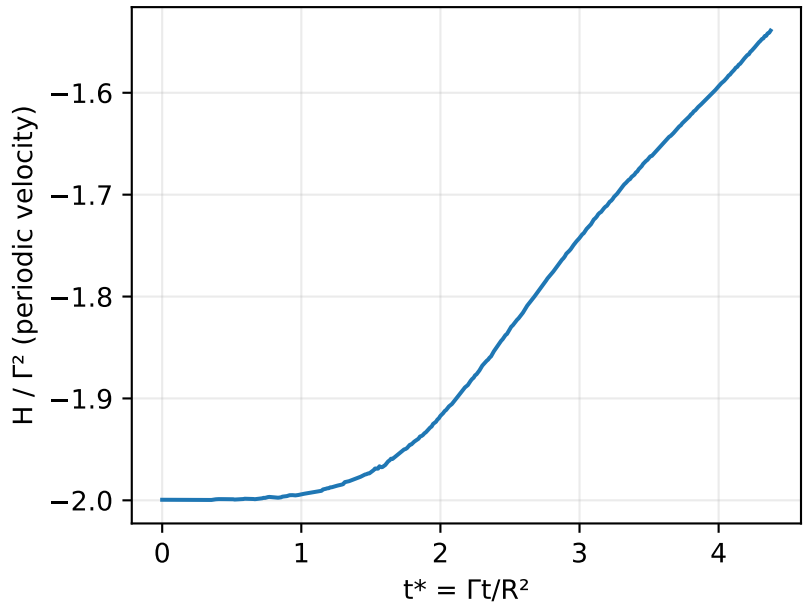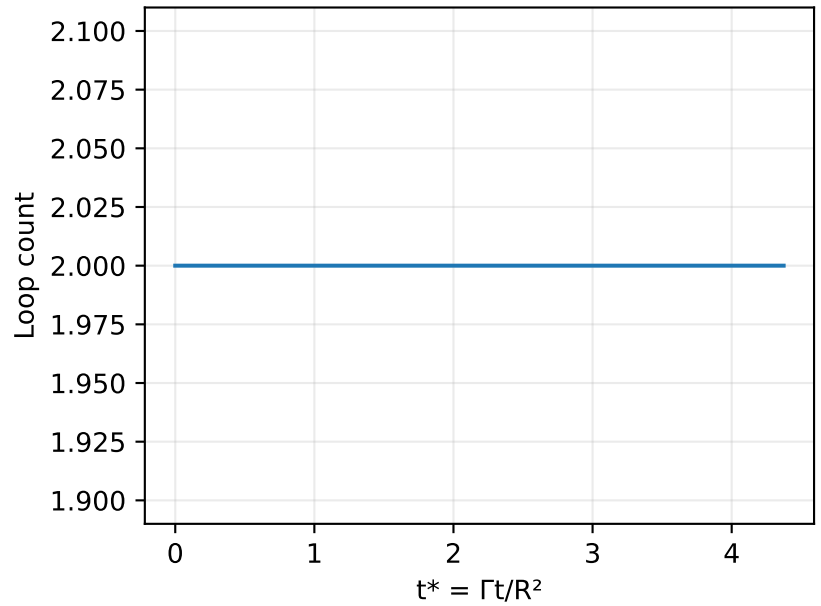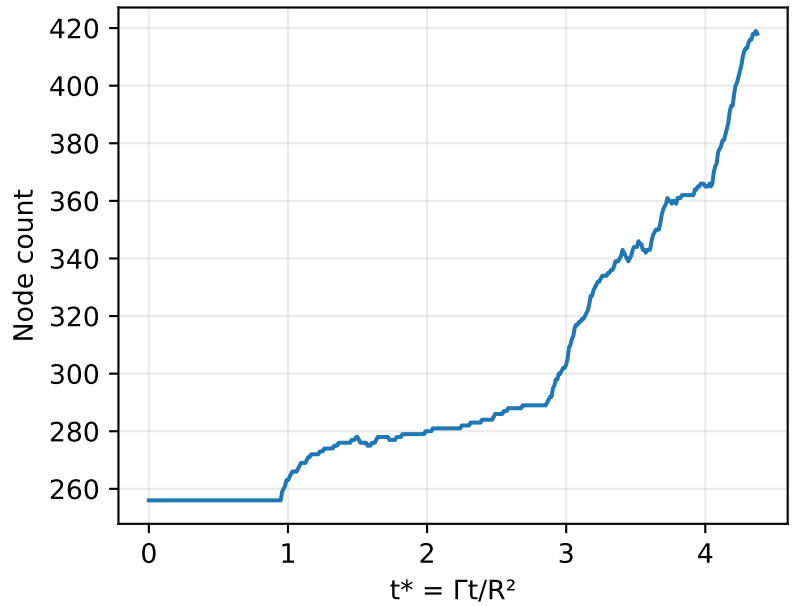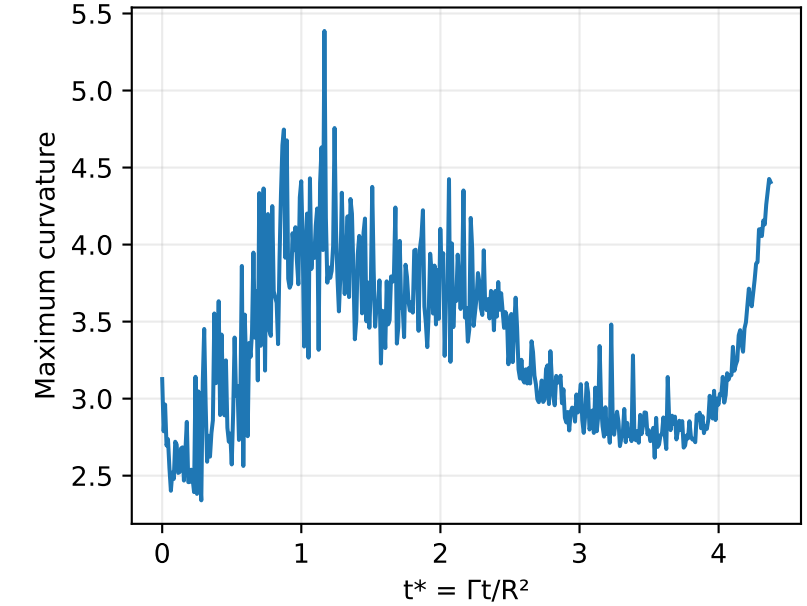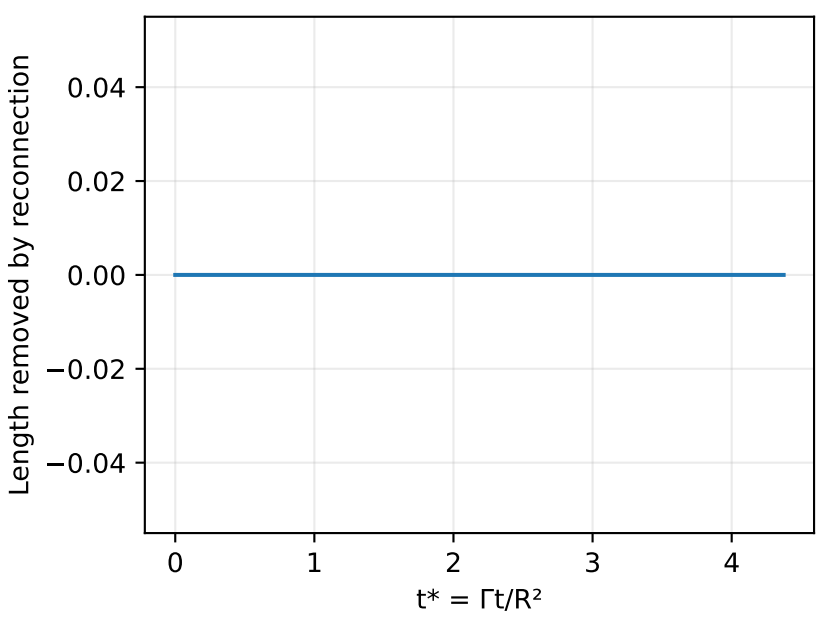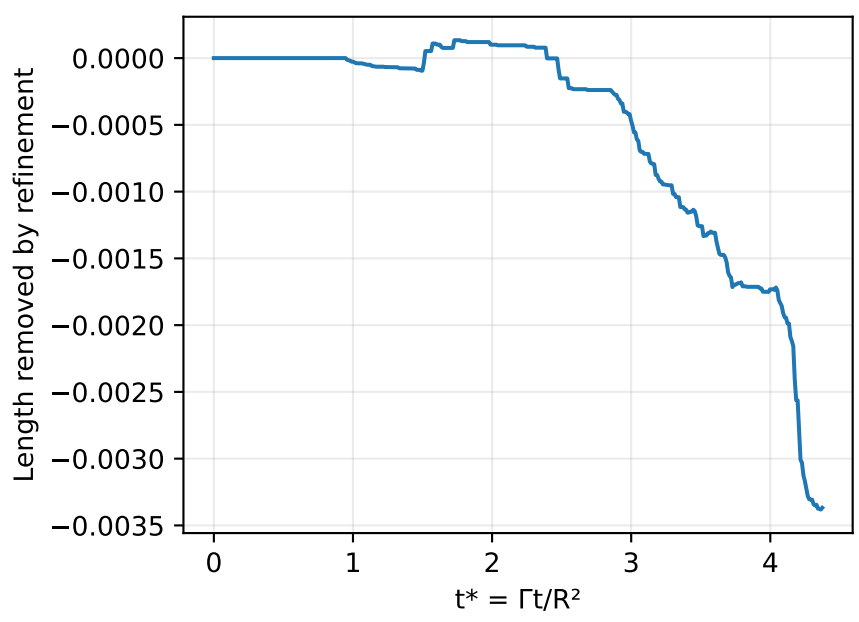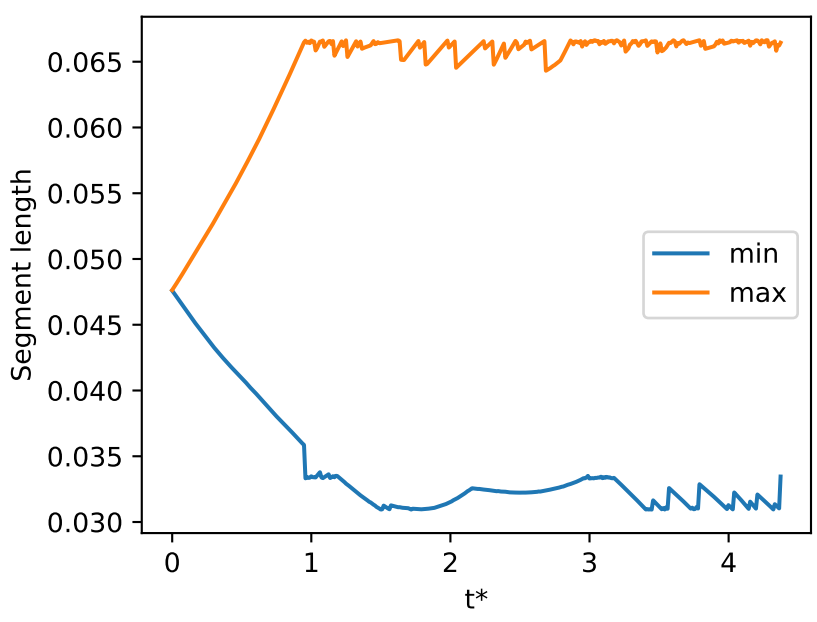hopf_prepared_fine · a=0.1171875 · no mutual friction